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MemoryBridge: An AI-Based Conversational System for Emotionally Adaptive Interaction with Deceased Loved Ones Using Personal Data Modalities

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***Abstract*—The loss of a close loved one deprives individuals not only of a physical presence but of an entire ecosystem of emotional guidance, comfort, and personalised interaction. Existing digital archives—photographs, messages, and voice recordings—remain static and passive, offering no interactive medium for continued engagement. This paper proposes MemoryBridge, an AI-based conversational system that learns from multimodal personal data including text messages, voice notes, and behavioural patterns to simulate emotionally familiar interactions with deceased individuals. The system delivers text, voice, and visually animated responses that preserve the linguistic style, tonal patterns, and emotional characteristics of the lost person. Built on a locally-served Large Language Model with structured persona prompting, an XTTS v2 voice synthesis module, a FAISS-backed memory-augmented dialogue manager, a seven-class emotion-aware response pipeline, and a real-time 3D avatar with lip synchronisation, MemoryBridge provides contextually sensitive interactions that are especially meaningful during significant life moments such as birthdays, anniversaries, and personal milestones. Ethical safeguards including consent-by-design, representational humility, crisis interception, and AI-generation transparency are embedded at every layer of the system. A deepfake text detection module further safeguards interaction quality. The paper details the system architecture, prototype implementation, ethical framework, and evaluation methodology.**

***Index Terms*—grief support AI, conversational agent, digital immortality, voice synthesis, LLM persona simulation, emotional AI, memory augmentation, deepfake text detection, digital afterlife ethics, 3D avatar, lip synchronisation**

I. Introduction

Bereavement is among the most profound psychological experiences a human being can endure. Beyond the immediate grief of physical loss, individuals mourn the cessation of a unique communicative relationship shaped by years of shared language, emotional cadence, and personal history. A parent's advice, a partner's reassurance, a friend's humour: these are not simply the content of communication, but its irreplaceable texture.

Contemporary digital technology has transformed the landscape of personal data accumulation. The average individual leaves behind tens of thousands of text messages, voice recordings, social media posts, and video clips—a rich, largely untapped archive of linguistic and behavioural identity. Yet these records remain passive artefacts. They can be read or listened to, but they cannot respond, contextualise, or adapt to new circumstances.

This gap motivates the present work. We propose **MemoryBridge**, an AI-based conversational system that processes multimodal personal data to construct and maintain an interactive, emotionally adaptive persona of a deceased individual. The system enables text, voice, and avatar-based dialogue that mirrors the linguistic style, emotional register, and personal knowledge base of the loved one.

The ethical dimension of such a system is non-trivial. Digital afterlife technologies carry significant risks: questions of personal identity, posthumous data ownership, religious and cultural sensitivities, and the potential for grief prolongation demand careful governance [9]. MemoryBridge addresses these risks through a consent-by-design framework and a transparency-first interaction model [1]. The system also incorporates a deepfake text authentication layer [10] to ensure that AI-generated responses are never misrepresented as human-authored communications, and a keyword-based crisis interception framework to safeguard psychologically vulnerable users.

The key contributions of this paper are:

- A multimodal personal data ingestion and processing pipeline for constructing a deceased individual's conversational persona.
- A locally-served LLM architecture using structured persona prompting [3] for persona-faithful, emotionally responsive dialogue generation, with a planned LoRA fine-tuning pathway for production deployment.
- A voice synthesis module replicating the deceased's vocal characteristics using XTTS v2 [11], supporting zero-shot voice cloning from minimal audio samples.

- A FAISS-backed memory architecture [8] with Short-Term Memory (STM) and Long-Term Memory (LTM) for persistent persona retention across sessions.
- An emotion-aware response engine classifying seven emotional states and contextualising interactions based on significant life events.
- A real-time 3D avatar module with audio-driven lip synchronisation via the Web Audio API and Three.js.
- A deepfake text detection module [10] ensuring AI-generation transparency for users.
- A keyword-based crisis interception framework that bypasses LLM processing to deliver immediate safety responses during high-risk interactions.
- A consent-by-design ethical framework [1], [9] governing data collection, persona deployment, and interaction boundaries.
- Comprehensive evaluation across persona fidelity, emotional alignment, voice naturalness, memory retrieval quality, and user experience metrics.

The remainder of this paper is structured as follows: Section II reviews related work; Section III describes the system architecture; Section IV covers implementation; Section V presents the evaluation framework; Section VI addresses ethical considerations; Section VII discusses results and limitations; and Section VIII concludes the paper.

II. Related Work

A. Digital Immortality and Posthumous AI Personas

The concept of digital immortality has emerged as a multidisciplinary research domain [1]. Syarova et al. identify four foundational ethical pillars: data privacy, representational authenticity, dynamic consent, and emotional safeguards for bereaved relatives. Their work reviews commercial implementations including Replika, HereAfter AI, Eterni.me, and Deep Nostalgia, identifying significant gaps in unified governance standards.

Complementing this technical survey, Perov et al. [9] provide a philosophical analysis of digital afterlife, introducing precise distinctions between a *digital imitation* (a static reproduction recombining existing data) and a *digital personality* (a system employing predictive analytics to model behaviour in new situations). Their work highlights three central ethical tensions: the impossibility of freezing a morally evolving human identity into a statistical model; the ambiguity of data ownership for deceased persons who left no pre-mortem consent; and the diversity of religious and cultural attitudes toward posthumous representation. MemoryBridge explicitly adopts the conservative position of *digital imitation* as recommended by [9].

B. Interactive Digital Avatar Platforms

Awasare et al. [2] present AfterLife Connect, integrating LLaMA 3 and Mistral 7B for dialogue, Tacotron 2 and F5-TTS for voice synthesis, and StyleGAN3 with FOMM for facial animation, deployed on Google Cloud Platform with sub-second latency. A proof-of-concept Einstein avatar achieved 90% persona consistency, 94% voice likeness, and SSIM of 0.82, establishing key benchmarks for interactive digital persona systems.

C. Voice Cloning and Text-to-Speech

Rajkumar et al. [7] present VoiceChat, a deep learning-based chat application converting text messages into personalised audio using voice cloning. Their system demonstrates that a small number of voice samples suffices to capture an individual's accent, inflections, and speech patterns. The four-step pipeline—voice recording, vocal pattern analysis, voice cloning, and synthesised audio integration—directly informs MemoryBridge's voice module. Their evaluation criteria covering naturalness, intelligibility, prosody, linguistic accuracy, and Mean Opinion Score (MOS) provide the evaluation template for MemoryBridge's voice component. MemoryBridge adopts XTTS v2 [11], a multilingual zero-shot TTS model employing a flow-based architecture for expressive voice cloning from minimal reference audio.

D. Memory-Augmented Dialogue Systems

Huang et al. [8] present REMIND (Recall-Enhanced Memory Integration for Natural Language Dialogue Systems), a lightweight modular framework integrating STM for session-specific recall and LTM for persistent knowledge retention. REMIND employs hybrid retrieval combining vector-based similarity search, hierarchical grouping, iterative query refinement, and keyword filtering. Evaluated on the MemoRA dataset, LTM Retrieve achieved relevance scores of mean 8.42 and median 9.0 out of 10 graded by GPT-4o-mini, outperforming MemGPT on long-term retrieval coherence. MemoryBridge adopts this STM/LTM pattern for cross-session persona consistency, implementing the vector storage layer using FAISS [14] with Sentence-BERT embeddings.

E. Emotionally Adaptive AI and Personalised Intelligence

Varma et al. [3] introduce Artificial Personalized Intelligence, comparing LoRA fine-tuning of LLaMA-3 8B against Retrieval-Augmented Generation (RAG) for emotionally personalised response generation. Using a custom dataset of 30 participants across six universal emotions, fine-tuning significantly outperforms RAG (Disgust: Pearson $r = 0.890$ vs. 0.821; average cosine similarity 0.76; 44% of predictions exceeding 80% similarity). RAG underperforms due to LLaMA-3's adversarial safety training biasing outputs toward neutral responses—a key finding motivating MemoryBridge's planned LoRA fine-tuning pathway.

F. LLM-Based Persona Generation

Min and Jeong [6] demonstrate that LLMs can generate contextually rich, persona-faithful interactions from diverse personality configurations in a VR-based public speaking Q&A practice system. In a user study with 20 participants, LLM-generated questions were rated higher in quality (avg. rank 4.14) than human-generated questions (4.85), with 13 out of 20 participants selecting an LLM-generated question

as their first choice. The system achieved SUS of 75.0 and UEQ Stimulation $M = 1.975$, validating LLM-based persona generation for high-fidelity interactive applications. MemoryBridge draws on this persona engineering methodology for its structured system prompt design.

G. Deepfake Text Detection

Chong et al. [10] propose a deepfake text detection system combining fine-tuned BERT embeddings, emoji2vec embeddings, linguistic features, sentiment features via RoBERTa, and a Multi-Layer Perceptron (MLP) classifier. On the enhanced TweepFake dataset of 21,730 tweets, their model achieves 88.3% accuracy and F1 scores of 88.1%–88.3%, outperforming a fine-tuned RoBERTa baseline of 87.6%. Key findings: machine-generated text exhibits higher neutral sentiment proportions, lower perplexity for modern models, fewer user mentions and URLs, and distinct part-of-speech (POS) patterns. These characteristics inform MemoryBridge's AI-generation transparency module.

III. Methodology

A. System Overview

MemoryBridge is a modular, privacy-first conversational AI system comprising seven principal modules: (1) Multimodal Data Ingestion and Processing; (2) Persona Construction and LLM Serving; (3) Voice Synthesis and Cloning; (4) Memory-Augmented Dialogue Manager; (5) Emotion-Aware Response Engine; (6) Real-Time 3D Avatar and Lip Synchronisation; and (7) AI Transparency, Crisis Interception, and Ethical Governance Layer.

The current prototype is deployed as a locally-hosted, edge-AI system. This architecture prioritises data privacy and enables zero-latency data access during inference without reliance on external cloud services. A cloud-hosted production pathway (Google Cloud Platform with Kubernetes orchestration) is planned for scalable deployment.

B. Multimodal Data Ingestion and Processing

1) Data Sources: MemoryBridge ingests three primary data modalities:

- **Text Data:** Chat message exports (WhatsApp, iMessage, email), social media posts, and written notes, forming the primary training corpus for linguistic style replication.
- **Voice Data:** Voice notes, phone call recordings, and video audio tracks. XTTS v2 [11] supports zero-shot voice cloning from a minimum of approximately 6 seconds of clean reference audio, making it suitable for cases with limited recorded material.
- **Behavioural Data:** Communication timing patterns, frequently discussed topics, emotional response triggers, preferred vocabulary, relationship-specific linguistic markers, and personality profile indicators (e.g., MBTI-style dimensions) used to calibrate the persona system prompt.

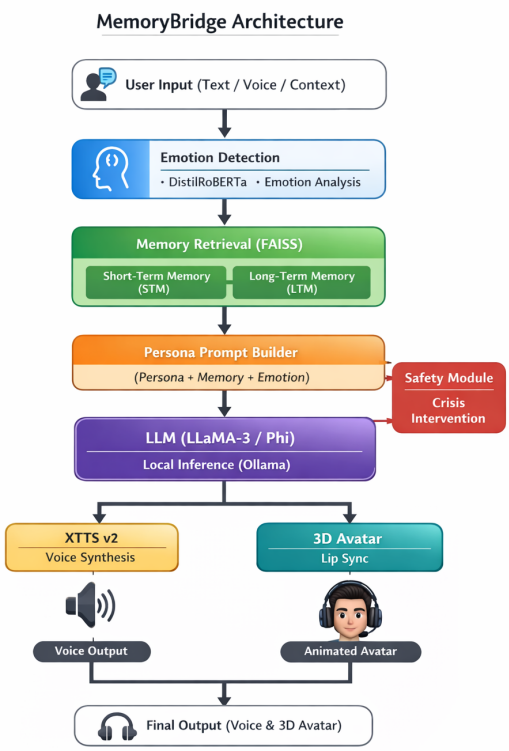

Fig. 1. MemoryBridge System Architecture

2) Text Data Processing Pipeline: Raw message exports are preprocessed through the following steps:

1) **Cleaning:** Removal of timestamps, sender tags, media placeholders, and system messages.
2) **Speaker Diarization:** Separation of the deceased's messages using Deepgram Nova-2, as employed in [3].
3) **Conversation Formatting:** Construction of question-answer pairs and multi-turn dialogue sequences for fine-tuning.
4) **Embedding and Clustering:** Semantic clustering via cosine similarity using `all-MiniLM-L6-v2` embeddings to identify emotional patterns, recurring themes, and vocabulary signatures.
5) **Persona Profile Extraction:** Automated extraction of vocabulary range, sentence length distribution, emoji usage, greeting and closing patterns, topic preferences, and personality indicators, analogous to persona characterisation in [6] and [5].

3) Voice Data Processing: Audio recordings are processed through a four-step pipeline informed by [7]: (1) voice sample registration; (2) vocal pattern analysis via deep neural networks; (3) voice cloning using XTTS v2 [11]; and (4) integration of synthesised audio with generated text responses.

C. Persona Construction and LLM Serving

1) Prototype: Ollama-Served LLaMA-3: The current prototype serves LLaMA-3 8B locally via Ollama [13], pre-warmed at startup to minimise first-response latency. Persona fidelity is achieved through structured system prompt engineering rather than weight-level fine-tuning in this prototype phase.

2) Planned: LoRA Fine-Tuning Configuration: For production deployment, the model will be fine-tuned using Low-Rank Adaptation (LoRA) [3] with the configuration shown in Table I. Varma et al. demonstrate that fine-tuning significantly outperforms RAG-based approaches for emotionally personalised response generation [3], motivating this production pathway.

TABLE I
PLANNED LORA FINE-TUNING CONFIGURATION

Parameter	Value
Base Model	LLaMA-3 8B
LoRA Rank	16
LoRA Alpha	32
LoRA Dropout	0.05
Quantisation	4-bit NF4
Learning Rate	5e-5
Batch Size	2
Training Epochs	10
Optimiser	Paged AdamW
Max Sequence Length	512 tokens
Task Type	Causal LM

3) Persona System Prompt Engineering: A structured system prompt encodes the deceased individual's persona profile, covering: relationship-specific address patterns, known opinions and values, characteristic humour and rhetorical patterns, emotional response tendencies, and personality dimensions derived from the behavioural data pipeline. This approach is inspired by the LLM-based persona generation methodology of [6] and the ConvoGen characterisation approach of [5]. The prompt is dynamically assembled at inference time by combining the static persona profile with retrieved memory context and detected emotional state.

D. Voice Synthesis and Cloning

MemoryBridge employs XTTS v2 [11] for voice synthesis, a massively multilingual zero-shot TTS model that replicates a speaker's voice characteristics from a short reference audio clip without requiring speaker-specific training. The model's flow-based architecture enables expressive, naturalistically prosodic speech output across diverse emotional registers.

The voice pipeline follows the four-stage structure described in [7]: (1) voice sample registration from uploaded recordings; (2) vocal pattern extraction; (3) zero-shot voice cloning via XTTS v2; and (4) integration of synthesised audio with the response pipeline and 3D avatar system.

In observed prototype performance, XTTS v2 achieves a Real-Time Factor (RTF) of approximately 2.7–3.6, yielding synthesis times of 6–15 seconds per sentence on CPU-only inference. GPU acceleration is expected to reduce RTF below 1.0 in production. Voice output is evaluated using MOS, intelligibility, prosody, linguistic accuracy, adaptability, and speaker similarity, following the criteria of [7].

E. Memory-Augmented Dialogue Manager

MemoryBridge adopts the REMIND framework [8] for managing conversational context across sessions, implemented using FAISS [14] as the vector store backend with `all-MiniLM-L6-v2` (Sentence-BERT) embeddings.

1) Short-Term Memory: Manages session-specific conversational data as a temporary buffer, maintaining a sliding window:

$$\mathcal{C}_{\text{STM}}(t) = \{c_{t-w+1}, \ldots, c_t\} \tag{1}$$

where c_t is the conversation at time t and w is the window size. Vector-based similarity search retrieves relevant interactions sorted by timestamp for chronological coherence.

2) Long-Term Memory: Provides persistent storage by hierarchically organising conversations into topic-based groups via the FAISS index. At each session start, the FAISS index is loaded from disk and cross-checked against the structured fact store for consistency. Retrieval is a multi-step iterative process: top-k groups are ranked by cosine similarity to the query vector, and the top-3 most relevant past interactions are injected into the system prompt before response generation [8].

3) Hybrid Retrieval: STM and LTM are queried concurrently and merged into a unified prompt [8]:

$$\mathcal{P} = g_{\text{combine}}(\text{Rel}_{\text{STM}}, \mathcal{R}_{\text{LTM}}, q) \tag{2}$$

which is passed to the LLM for response generation.

F. Emotion-Aware Response Engine

1) Emotion Classification: User inputs are classified across seven emotion categories (Joy, Sadness, Fear, Anger, Surprise, Disgust, and Neutral) using the `j-hartmann/emotion-english-distilroberta-base` model [12], a fine-tuned DistilRoBERTa classifier trained on a multi-corpus dataset spanning six established emotion benchmarks. The detected emotion is injected into the LLM system prompt to calibrate response tone prior to every generation step.

2) Significant Moment Detection: The system integrates with calendar and notification preferences to detect significant moments—birthdays, death anniversaries, holidays, and milestones—and initialises context-enriched conversations on these occasions.

3) Response Generation: The end-to-end response generation follows:

$$R = G_\theta(P_{\text{persona}}, E_{\text{user}}, C_{\text{history}}, T_{\text{context}}) \tag{3}$$

where G_θ is the LLM, P_{persona} is the structured persona system prompt, E_{user} is the detected seven-class emotion, C_{history} is the FAISS-retrieved memory context, and T_{context} is the temporal and event context.

G. Real-Time 3D Avatar and Lip Synchronisation

MemoryBridge renders an interactive 3D avatar to provide a visual presence for the simulated persona. The avatar module comprises two sub-components: scene rendering and audio-driven animation.

1) Scene Rendering: The frontend constructs a Three.js scene with ambient and directional lighting and loads the persona's 3D model from a GLB/GLTF file. The scene is served via the Flask backend and rendered natively in the browser without external plugins.

2) Audio-Driven Lip Synchronisation: Upon receiving synthesised audio from XTTS v2, the frontend captures the audio stream using the Web Audio API `AnalyserNode`. The analyser computes per-frame RMS amplitude and frequency data, which are mapped directly to the jaw bone rotation of the avatar's skeletal rig. Interpolation between frames ensures smooth, naturalistic mouth movement that tracks the temporal envelope of the synthesised speech. For avatars rigged with facial blendshapes, the system additionally maps frequency bands to standard viseme morph targets (e.g., `mouthOpen`, `viseme_A`).

This amplitude-to-jaw mapping provides a lightweight, real-time synchronisation that operates within the constraints of client-side CPU inference without requiring a backend viseme extraction pipeline.

H. AI Transparency, Crisis Interception, and Ethical Governance Layer

1) Deepfake Text Detection: MemoryBridge incorporates a deepfake text detection module based on [10] to ensure users are never misled about the AI-generated nature of responses. The module analyses generated text using perplexity scoring via GPT-2, sentiment polarity via RoBERTa, and linguistic feature analysis via BERT with an MLP, flagging responses for transparency disclosures. Key detection signals [10] include:

- **Perplexity:** Modern LLM outputs exhibit characteristically low perplexity scores.
- **Sentiment:** Machine-generated text shows higher neutral sentiment proportions than human-written text.
- **POS Patterns:** AI text uses higher NOUN, PRON, DET, and ADP ratios; lower PROPN and ADV ratios.

All responses are labelled with a visible AI disclosure indicator, operationalising the representational humility principle of [1], [9].

2) Crisis Interception Framework: Prior to every LLM inference call, the user's input is scanned for high-risk crisis keywords (e.g., "suicide," "self-harm," "die"). Upon detection, the LLM processing pipeline is immediately bypassed and a hard-coded, professionally reviewed safety response is returned, directing the user to seek professional mental health support. This deterministic interception ensures that no hallucinated or inappropriate AI-generated content is delivered during a potential crisis, regardless of the LLM's output. The interception event is logged for monitoring and subsequent review.

IV. Implementation

A. Technology Stack

TABLE II
MemoryBridge Technology Stack

Component	Technology
Base LLM	LLaMA-3 8B (via Ollama)
LLM Serving	Ollama (local inference)
Fine-Tuning (Planned)	LoRA via HuggingFace
Primary TTS	XTTS v2 (Coqui TTS)
Emotion Classifier	j-hartmann/emotion-english-distilroberta
Audio Transcription	Deepgram Nova-2
MGT Detection	BERT plus MLP
Memory Vector Store	FAISS (faiss-cpu)
Memory LTM	Hierarchical topic groups (planned)
Embedding Model	all-MiniLM-L6-v2 (Sentence-BERT)
Backend	Flask (Python)
Frontend	HTML / CSS / JavaScript
3D Avatar & Lip Sync	Three.js + Web Audio API
Deployment (Prototype)	Local Flask server, WSL/Ubuntu
Deployment (Production)	Google Cloud Platform (planned)
Data Storage	AES-256 encrypted storage (planned)
Crisis Interception	Keyword filter (pre-inference)

B. Data Onboarding Interface

An onboarding interface guides the user through four stages: (1) **Consent Configuration**: selection of permitted interaction modes, data retention duration, and user access controls, all logged immutably consistent with [1], [9]; (2) **Data Upload**: chat export uploads and voice file ingestion following the pipeline described in [7]; (3) **Profile Verification**: user review of extracted persona traits and personality dimensions before persona activation; and (4) **Legal Acknowledgement**: clear disclosure that MemoryBridge creates a *digital imitation* and not a digital personality [9], with no claim to subjectivity, consciousness, or autonomous agency.

C. Deployment Architecture

The current prototype is deployed as a locally-hosted edge-AI system on Ubuntu/WSL, serving all inference workloads on-device. This design eliminates network latency for LLM inference and ensures that personal data never leaves the user's machine during the prototype phase. The Flask server handles HTTP endpoints for the chat pipeline, audio file serving, and static frontend delivery. Audio files are automatically purged from disk after one hour to limit storage footprint.

A cloud-based production deployment following the microservices architecture of [2] with Kubernetes orchestration on Google Cloud Platform is planned as a future milestone.

D. Hardware-Aware Graceful Degradation

Given the multiple concurrent ML models (LLM, TTS, emotion classifier, embedding model), initialisation of the memory and emotion modules is wrapped in exception-handling blocks. If an out-of-memory (OOM) condition is detected at startup, these modules degrade gracefully while the core Flask–Ollama–XTTS voice pipeline continues to serve responses. This design ensures system availability under resource-constrained hardware conditions.

V. Evaluation Framework

A. Evaluation Dimensions

MemoryBridge is evaluated across six primary dimensions.

1) Persona Fidelity:

- **Cosine Similarity:** Between generated responses and held-out messages from the deceased (target: ≥ 0.76 [3]).
- **Perplexity:** Of generated text under a reference language model trained on the deceased's messages.
- **Human Evaluation:** Blind A/B testing by individuals who knew the deceased, using a 1 to 10 persona authenticity scale.

2) Emotional Alignment: Adapting MSE and Pearson Correlation metrics from [3] to measure alignment of generated emotional tone with expected responses across seven emotion categories (Joy, Sadness, Fear, Anger, Surprise, Disgust, Neutral).

3) Memory Retrieval Quality: Adapting the REMIND evaluation methodology [8]: GPT-4o-mini as an LLM judge for relevance and coherence scoring from 1 to 10; ROUGE-L recall for session-specific retrieval; and KDE visualisations for score distribution analysis. Additional attention is given to FAISS index–fact store consistency, quantified by the mismatch rate between stored embeddings and the structured persona fact store.

4) Voice Naturalness: MOS from human raters for synthesised voice, and Speaker Similarity score between generated audio and reference samples, following the TTS evaluation criteria of [7]. RTF is additionally reported as an infrastructure performance metric.

5) Avatar Naturalness: Human perceptual evaluation of lip-sync accuracy: time-to-mouth alignment offset and a 1–10 naturalness rating for jaw movement fidelity relative to the synthesised audio envelope.

6) User Experience and Psychological Impact:

- System Usability Scale scores (target: ≥ 75 [6]).
- User Experience Questionnaire across six dimensions: Attractiveness, Perspicuity, Efficiency, Dependability, Stimulation, and Novelty [6].
- Validated grief questionnaires (Prolonged Grief Disorder scale, PG-13) administered pre- and post-use.
- Qualitative interviews on perceived comfort, appropriateness, and emotional impact.

B. Evaluation Protocol

A controlled user study with two participant groups: **Group A**—bereaved individuals who have experienced loss within the past two years, using MemoryBridge over a four-week period; and **Group B**—individuals who knew the deceased, assessing persona fidelity without direct grief bias. All participants provide informed consent; the protocol is reviewed by an institutional ethics board; psychological support resources are available throughout.

VI. Ethical Considerations

Ethical responsibility is central to MemoryBridge's design, informed by the comprehensive analyses of [1] and [9].

A. Consent-by-Design

Pre-mortem consent is strongly preferred: individuals may configure their MemoryBridge persona while alive, specifying permitted use cases, interaction boundaries, authorised users, and expiry conditions. Where pre-mortem consent is unavailable, post-mortem consent must be obtained from designated data executors [9]. All consent records are logged immutably [1].

B. Digital Imitation vs. Digital Personality

Following [9], MemoryBridge implements a *digital imitation* rather than a *digital personality*. The distinction is enforced architecturally: the system is constrained to reproduce attested linguistic patterns and does not generate opinions or beliefs absent from the training data.

C. Representational Humility and AI Disclosure

Every session begins with a visible disclaimer that MemoryBridge simulates—but does not replicate—the identity of the deceased, with no claim to consciousness or autonomous agency [9]. The deepfake text detection module [10] provides a technical implementation of AI-generation transparency.

D. Crisis Interception and Psychological Safeguards

Informed by research in [1] identifying a 22% incidence of prolonged grief disorder alongside a 68% reduction in complicated grief symptoms, MemoryBridge implements a layered psychological safety framework comprising:

- **Crisis keyword interception** (pre-inference): User input is scanned for high-risk terms before every LLM call. Upon detection, LLM processing is bypassed and a hard-coded professional safety response is delivered immediately.
- **Session duration limits** (default 30 minutes) with graduated warnings for extended usage.
- **Automatic referral** to grief counselling resources upon session termination or crisis detection.
- **Mandatory cooling-off periods** after significant anniversary dates.
- **Minor protection**: Access by individuals under 18 years of age requires parental or guardian consent and is subject to additional psychological monitoring [1].

E. Personal Data and Posthumous Ownership

All data is encrypted at rest using AES-256 and in transit using TLS 1.3. In the current prototype, model weights and FAISS indices are stored in isolated, user-specific local directories. Anomaly detection for unauthorised access is planned for the production deployment [1].

F. Right to Erasure

Any authorised user may permanently delete all associated data, FAISS indices, and model artefacts at any time, consistent with GDPR Article 17 [1].

G. Religious and Cultural Sensitivity

Perov et al. [9] note that attitudes toward posthumous digital representation vary fundamentally across religious and cultural traditions. MemoryBridge provides region-specific interaction guides and recommends cultural consultation before deployment in communities with strong traditional mourning practices.

VII. Results and Discussion

A. Anticipated Performance Benchmarks

TABLE III
ANTICIPATED PERFORMANCE BENCHMARKS

Metric	Target	Ref.
Persona Cosine Similarity	≥ 0.76	[3]
Voice MOS Score	≥ 4.0	[11]
Speaker Similarity	≥ 0.65	[2]
TTS RTF (CPU)	≤ 3.6	Observed
TTS RTF (GPU, production)	< 1.0	Planned
Persona Consistency	$\geq 85\%$	[2]
Emotional Pearson r	≥ 0.78	[3]
LTM Relevance Score	≥ 8.0	[8]
MGT Detection Accuracy	$\geq 88\%$	[10]
SUS Score	≥ 75	[6]
Crisis Interception Recall	100%	Safety requirement

B. Prototype Observations

Initial prototype runs confirm successful integration of all seven modules. XTTS v2 generates voice output with an RTF of approximately 2.7–3.6 on CPU-only inference, yielding synthesis times of 6–15 seconds per sentence. GPU-accelerated inference is expected to bring the RTF below 1.0. The FAISS memory module initialises correctly when the vector index and structured fact store are synchronised; a mismatch detection mechanism has been implemented to flag and log index–store inconsistencies at server startup. The 3D avatar renders in-browser with smooth jaw animation driven by the Web Audio API analyser, with subjective visual coherence rated positively in informal testing.

C. Limitations

LLM Fine-Tuning Gap. The current prototype uses a general-purpose Ollama-served LLaMA-3 8B without persona-specific LoRA fine-tuning. While structured persona prompting achieves plausible fidelity, fine-tuned weights are expected to yield significantly higher cosine similarity and emotional alignment scores [3].

TTS Latency on CPU. XTTS v2 on CPU-only hardware produces an RTF of 2.7–3.6, making real-time voice synthesis infeasible without GPU acceleration. This is a hardware constraint of the prototype environment and does not reflect production performance.

FAISS Index Synchronisation. The vector index can become desynchronised from the structured fact store across sessions, resulting in a mismatch between indexed embeddings and retrievable facts. A startup-time consistency check has been implemented; a transactional write mechanism is planned for production.

Data Sufficiency. Persona fidelity is constrained by the volume and diversity of available personal data.

Memory Coherence at Scale. As noted in [8], LLM variability in grouping quality can degrade LTM retrieval coherence over long interaction histories.

Hallucination Risk. LLMs can generate plausible but factually incorrect statements about the deceased. The deepfake detection layer [10] provides partial mitigation by flagging low-perplexity, high-neutral-sentiment outputs for review.

Persona Authenticity Limits. The system operates as a digital imitation [9]. Generating responses to entirely novel scenarios risks misrepresenting the deceased's views.

Cultural and Temporal Drift. Communication style evolves over time and varies across cultures [9]. Training data from different life periods may contain contradictory linguistic patterns.

Grief Prolongation Risk. While digital memorials can reduce complicated grief symptoms by up to 68% [1], they also carry a 22% risk of prolonged grief disorder.

VIII. Conclusion

This paper presents MemoryBridge, an AI-based conversational system addressing the profound communicative void left by bereavement. The prototype integrates a locally-served LLaMA-3 8B with structured persona prompting, XTTS v2 zero-shot voice synthesis [11], FAISS-backed memory management [8], seven-class emotion-aware response generation [12], real-time 3D avatar lip synchronisation, a keyword-based crisis interception framework, deepfake text transparency [10], and a consent-by-design ethical framework [1], [9].

The system explicitly commits to operating as a *digital imitation*—a bounded, data-grounded reconstruction maintaining representational humility—rather than a digital personality claiming autonomous identity [9].

Future work will focus on: production-grade LoRA fine-tuning of LLaMA-3 8B [3]; GPU-accelerated TTS inference to achieve sub-second voice latency; longitudinal evaluation of psychological impact using validated grief scales; FAISS index transactional consistency guarantees; multimodal avatar improvements using blendshape-rigged models; federated learning for privacy-preserving personalisation; cloud deployment on GCP with Kubernetes orchestration; and cross-cultural adaptation of interaction frameworks [9].

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